Zach Heckert | Software Engineer

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Primary Technical Expertise

JavaScript (ES6+), TypeScript, React.js, Svelte/SvelteKit, HTML5/CSS3, Tailwind, Ruby, Rails, Node.js, Express.js, Vite, MongoDB, Postman, Git, GitHub, GitHub Actions (CI/CD), Docker, Netlify, Render, Heroku, AWS (S3), Figma, Jira/Confluence, Google Suite, AI API Integration, Adobe Photoshop/Illustrator

Experience

Software Engineer 1 - ExecOnline, Inc August 2022 - January 2024

- Developed **type-safe API's** for an enterprise-scale ed-tech app bridging backend and frontend using functional programming principles, Decoders, MobX, React, TypeScript, Ruby on Rails and a custom Ruby DSL, **reducing frontend errors by 100%** and enhancing system reliability.
- Implemented **frontend UI for consumer-facing features** as part of a company-wide shift to a SaaS-based business model, significantly enhancing user experience through intuitive design and advanced customization options, **contributing to an 85% retention rate** driven by improved user engagement and satisfaction.
- Proactively cataloged, triaged, and resolved hundreds of bug alerts using Honeybadger, **mitigating recurring issues and silencing 10,000+ noisy alerts**, ensuring a smoother developer experience.
- Rebuilt a critical portion of the company's core project, earning trust from key collaborators through clear communication and dependable delivery; **recognized with a company award for fostering a collaborative and efficient work environment**.

Associate Software Engineer - ExecOnline, Inc May 2021 - August 2022

- Partnered with the Director of UX to lead and document a **comprehensive platform-wide branding refresh**. Tasks included updating, consolidating, or removing SASS stylesheets, refactoring and updating CSS variables and their references, replacing fonts, imagery, logos, and various other styling elements and patterns throughout the application, **resulting in more manageable CSS, ensuring both smooth future development and enhanced brand recognition**.
- **Contributed to achieving AA accessibility status** for our product by optimizing tags and validating content functionality with tools like Voiceover, **ensuring an inclusive user experience**.
- Improved performance and scalability by modernizing key client-facing pages. **Replaced legacy HAML views with React components and TypeScript reducing load times by over 100%**, preventing hard reloads, ensuring seamless user experiences for 10,000+ active users.

Current Projects

Full-Stack Note App - https://github.com/zheckert/pkms
Technologies used: Ruby on Rails, React

- Demonstrated adaptability by identifying and **resolving challenges with session-based authentication** in an API-only Rails backend, **currently transitioning to a token-based approach with JWT to ensure secure and scalable user sessions**.
- **Leveraged ActiveRecord methods like joins, includes, and distinct** to optimize database queries, reducing redundant lookups and improving the performance of complex relationships like notes and tags.
- **Developed a custom filtering endpoint that dynamically fetches notes by tags**, showcasing proficiency in designing intuitive features and efficiently handling many-to-many relationships in Rails.
- **Built modular React components with attention to UX and state management**, including a dynamic notes creation form, a sortable and filterable notes list, and actionable error-handling for seamless frontend-backend interaction.

SvelteKit URL Shortener - https://github.com/zheckert/cute-url-shortener-api
Technologies used: SvelteKit, MongoDB

- **Developed a URL shortening API using SvelteKit and MongoDB**, enabling users to generate compact, shareable links with custom base62 encoding for improved readability and uniqueness.
- **Implemented server-side validations to ensure data integrity**, including checks for URL structure, length, and duplicate entries, reducing potential user errors and improving system reliability.
- **Enhanced user experience with rate-limiting middleware** to prevent abuse while ensuring fair usage for clients.
- **Designed a dynamic environment-based base URL system** to seamlessly support development and production environments, improving deployment flexibility and scalability.

Education

Full-Stack JavaScript Web Development Certification V School
Bachelor of Science, Art and Visual Communication Utah Valley University
Associate in Applied Science, Digital Communication Technology Utah Valley University